

Canary Rollout Prediction Shift Analysis

Detecting and Responding to ML Model Deployment Issues

Objective

Learn to identify, diagnose, and respond to prediction distribution shifts during canary deployments.

A model that runs without errors is NOT necessarily a model that works.

Problem Statement: The Scenario

Situation

- bullet New classifier model deployed via canary rollout (10% traffic)
- bullet After 2 hours: Class A predictions shift from 20% -> 55%

Observations

- bullet Latency: Normal ■
- bullet Error Rate: Normal ■
- bullet Ground Truth Labels: NOT available ■

Challenge: The model is acting weird, but technically 'healthy'.

Real-World Relevance

Industry Examples

- bullet Finance: Fraud model flags 3x more transactions (User Friction)
- bullet Healthcare: Diagnosis model becomes over-sensitive (Patient Harm)
- bullet E-commerce: Recommender shows only one category (Revenue Loss)

Why it Matters

- bullet Silent failures destroy user trust
- bullet You can't wait for labels to fix bad predictions

Input Data Overview

The Datasets

- bullet Baseline: 10,000 samples from stable/old model
- bullet Canary: 10,000 samples from new model

Key Features

- bullet feature_1: Numerical (Normal dist, mean~50)
- bullet feature_2: Numerical (Normal dist, mean~100)
- bullet feature_3: Uniform (0-1)

Core Concepts (High Level)

The Toolkit

- bullet Canary Rollout: Gradual deployment (10% -> 100%)
- bullet Prediction Drift: When model outputs change
- bullet Covariate Shift: When model INPUTS change
- bullet PSI: Population Stability Index
- bullet KS-Test: Statistical distribution comparison

Concepts Breakdown (Simple)

Definitions

- bullet PSI > 0.25 : 'Stop the presses, something is wrong'
- bullet Covariate Shift: 'Teacher grading exams from a different school'
- bullet Prediction Drift: 'Doctor diagnosing everyone with the same flu'

Statistical tests allow us to see 'invisible' problems.

Step-by-Step Solution Flow

- bullet 1. Simulate predictions for Baseline & Canary
- bullet 2. Check Data Quality (Nulls, Types)
- bullet 3. Detect Input Drift (PSI, KS-Test)
- bullet 4. Analyze Prediction Behavior (Chi-Square)
- bullet 5. Calculate Risk Signals
- bullet 6. Recommend Safest Action

Code Logic Summary

Key Functions

- bullet `simulate_baseline_predictions()`: Generate stable reference data
- bullet `calculate_psi()`: The math logic for drift detection
- bullet `detect_input_drift()`: Runs PSI + KS-test on all columns
- bullet `determine_safest_action()`: Logic tree for decision making

PSI: Important Function

calculate_psi(expected, actual)

- bullet Bins the data into 10 buckets
- bullet Compares % of counts in each bucket
- bullet Returns a single score

PSI < 0.1 (OK) | 0.1-0.25 (Monitor) | >0.25 (Action Required)

Execution Output

The Results

- bullet feature_1 PSI: 0.32 ■ (Significant Input Drift)
- bullet Class A Shift: +35% ■■ (Major Prediction Drift)
- bullet Data Quality: Clean ■

Verdict

The model is predicting differently because the INPUTS changed.

Observations & Insights

What happened?

bullet feature_1 mean shifted from 50 -> 65

bullet The model learned 'Higher feature_1 = Class A'

bullet Thus, more Class A predictions

This is a classic case of Covariate Shift causing Prediction Drift.

Advantages & Limitations

Advantages

- bullet Fast detection (don't need to wait days for labels)
- bullet Objective, math-based decision making

Limitations

- bullet Doesn't tell you if the new predictions are WRONG, just DIFFERENT
- bullet Requires careful threshold tuning

Interview Key Takeaways

Q: What if latency is fine but predictions change?

bullet A: Check Input Drift (PSI) first.

Q: When to Rollback?

bullet A: If $PSI > 0.25$ or prediction shift $> 10\%$ without explanation.

Q: Why Canary?

bullet A: To catch these exact issues on only 10% of users.

Conclusion

Final Recommendation

PAUSE + ROUTE TO REVIEW

Why?

- bullet We have confirmed Input Drift.
- bullet We cannot trust the model output.
- bullet Safety First: Investigate feature_1 upstream data source.